

161. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a non constant twice differentiable function such that $g'\left(\frac{1}{2}\right) = g'\left(\frac{3}{2}\right)$. If a real valued function

f is defined as $f(x) = \frac{1}{2}[g(x) + g(2-x)]$, then :

(A) $f''(x) = 0$ for no x in $(0, 1)$

(B) $f''(x) = 0$ for exactly one x in $(0, 1)$

(C) $f'\left(\frac{3}{2}\right) + f'\left(\frac{1}{2}\right) = 1$

(D) $f''(x) = 0$ for atleast two x in $(0, 2)$

माना $g : \mathbb{R} \rightarrow \mathbb{R}$ एक परिवर्तनीय तथा दो बार अवकलनीय फलन है और $g'\left(\frac{1}{2}\right) = g'\left(\frac{3}{2}\right)$ है। यदि एक वास्तविक मान फलन f ,

$f(x) = \frac{1}{2}[g(x) + g(2-x)]$ द्वारा परिभाषित है, तो :

(A) $(0,1)$ में कोई भी x नहीं है, जिसके लिए $f''(x) = 0$ है

(B) $(0,1)$ में ठीक एक x के लिए $f''(x) = 0$ है

(C) $f'\left(\frac{3}{2}\right) + f'\left(\frac{1}{2}\right) = 1$

(D) $(0,2)$ में कम से कम दो x के लिए $f''(x) = 0$ है

Ans. D

Sol. $f'(x) = \frac{1}{2}(g'(x) - g'(2-x))$

Since $f'\left(\frac{1}{2}\right) = f'(1) = f'\left(\frac{3}{2}\right) = 0$ (Rolle Theorem)

$\Rightarrow f''(x) = 0$ for atleast 2 values of $x \in (0,2)$

162. If the function $f : (-\infty, -1] \rightarrow (a, b]$ defined by $f(x) = e^{x^3-3x+1}$ is one – one and onto, then the distance of the point $P(2b+4, a+2)$ from the line $x + e^{-3}y = 4$ is :

यदि $f(x) = e^{x^3-3x+1}$ द्वारा परिभाषित फलन $f : (-\infty, -1] \rightarrow (a, b]$ एकैकी तथा आच्छादी है, तो रेखा $x + e^{-3}y = 4$ से बिन्दु $P(2b+4, a+2)$ की दूरी है :

(A) $\sqrt{1+e^6}$

(B) $2\sqrt{1+e^6}$

(C) $4\sqrt{1+e^6}$

(D) $3\sqrt{1+e^6}$

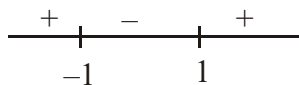
Ans. B

Sol. $f : (-\infty, -1] \rightarrow (a, b]$

$f(x) = e^{x^3-3x+1}$

$f'(x) = e^{x^3-3x+1}(3x^2-3)$

$$f'(x) = e^{x^3-3x+1} 3(x-1)(x+1)$$



$$(-\infty, -1] \rightarrow \uparrow F^n$$

$$\text{So } x \rightarrow -\infty$$

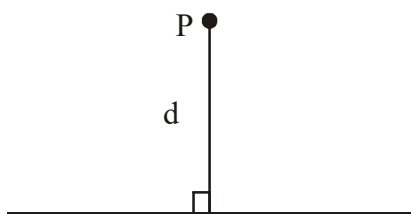
$$\Rightarrow f(x) = e^{-\infty} \rightarrow 0 = a$$

$$x \rightarrow -1$$

$$\Rightarrow F(-1) = e^{-1+3+1} = e^3 = b$$

$$(a, b] \Rightarrow (0, e^3]$$

$$P(2e^3 + 4, 2)$$



$$d = \left| \frac{2e^3 + 4 + e^{-3}(2) - 4}{\sqrt{1 + (e^{-3})^2}} \right|$$

$$d = \frac{2(e^3 + e^{-3})}{\sqrt{\frac{e^6 + 1}{e^6}}}$$

$$d = \frac{\frac{2}{e^3}(1 + e^6)}{\frac{\sqrt{1 + e^6}}{e^3}}$$

$$d = 2\sqrt{1 + e^6}$$

163. If $5f(x) + 4f\left(\frac{1}{x}\right) = x^2 - 2$, $\forall x \neq 0$ and $y = 9x^2 f(x)$, then y is strictly increasing in :

यदि $5f(x) + 4f\left(\frac{1}{x}\right) = x^2 - 2$, $\forall x \neq 0$ है तथा $y = 9x^2 f(x)$ है, तो किस अन्तराल में y निरंतर वर्धमान है :

(A) $\left(0, \frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$

(B) $\left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(0, \frac{1}{\sqrt{5}}\right)$

(C) $\left(-\infty, \frac{1}{\sqrt{5}}\right) \cup \left(0, \frac{1}{\sqrt{5}}\right)$

(D) $\left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$

Ans. D

Sol. $5f(x) + 4f\left(\frac{1}{x}\right) = x^2 - 2 \quad \dots(1)$

Substitute $x \rightarrow \frac{1}{x}$

$5f\left(\frac{1}{x}\right) + 4f(x) = \frac{1}{x^2} - 2 \quad \dots(2)$

$(1) \times 5 - (2) \times 4$

$\Rightarrow 9f(x) = 5x^2 - 10 - \frac{4}{x^2} + 8$

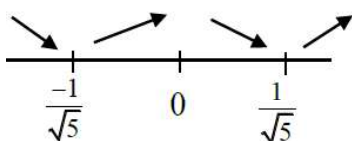
$\Rightarrow 9f(x) = \frac{5x^2 - 2x^2 - 4}{x^2}$

$\Rightarrow 9x^2 f(x) = 5x^4 - 2x^2 - 4$

$y = 5x^4 - 2x^2 - 4$

$\frac{dy}{dx} = 20x^3 - 4x > 0$ (given)

$\Rightarrow 4x(5x^2 - 1) > 0$



$\Rightarrow y$ is increasing in $\left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$

164. Let $f : [2, 4] \rightarrow \mathbb{R}$ be a differentiable function such that

$(x \log_e x) f'(x) + (\log_e x) f(x) + f(x) \geq 1, x \in [2, 4]$ with $f(2) = \frac{1}{2}$ and $f(4) = \frac{1}{4}$.

Consider the following two statements :

(A) : $f(x) \leq 1$, for all $x \in [2, 4]$

(B) : $f(x) \geq \frac{1}{8}$, for all $x \in [2, 4]$

Then :

(A) Only statement (A) is true

(B) Both the statements (A) and (B) are true

(C) Neither statement (A) nor statement (B) is true

(D) Only statement (B) is true

माना $f : [2, 4] \rightarrow \mathbb{R}$ एक अवकलनीय फलन है, जिसके लिए

$$(x \log_e x) f'(x) + (\log_e x) f(x) + f(x) \geq 1, x \in [2, 4], f(2) = \frac{1}{2} \text{ तथा } f(4) = \frac{1}{4} \text{ है।}$$

निम्न में दो कथनों का विचार कीजिए :

(A) : सभी $x \in [2, 4]$ के लिए $f(x) \leq 1$ है।

(B) : सभी $x \in [2, 4]$ के लिए $f(x) \geq \frac{1}{8}$ है।

तो :

(A) केवल कथन (A) सत्य है

(B) (A) तथा (B) दोनों कथन सत्य हैं

(C) न तो कथन (A) न ही कथन (B) सत्य है

(D) केवल कथन (B) सत्य है

Ans. B

Sol. $\frac{d}{dx}(x \ln(x) \cdot f(x) - x) \geq 0, \forall x \in [2, 4]$

$$\text{let } g(x) = x \ln(x) f(x) - x, \quad \forall x \in [2, 4]$$

$$\Rightarrow g'(x) \geq 0, \forall x \in [2, 4]$$

$$\Rightarrow g'(x) \text{ is increasing function in } [2, 4]$$

$$\Rightarrow g(2) \leq g(x) \leq g(4)$$

$$\Rightarrow 2 \cdot \ln 2 \cdot \frac{1}{2} - 2 \leq x \ln(x) f(x) - x \leq 4 \cdot \ln(4) \cdot \frac{1}{4} - 4$$

$$\Rightarrow \ln 2 - 2 + x \leq x \cdot \ln x \cdot f(x) \leq 2(\ln 2 - 2) + x$$

$$\Rightarrow \frac{\ln 2 - 2}{x \ln x} + \frac{1}{\ln x} \leq f(x) \leq \frac{2(\ln 2 - 2)}{x \ln x} + \frac{1}{\ln x}$$

Since numerator is constant

$$\Rightarrow \frac{\ln 2 - 2}{4 \ln 4} + \frac{1}{\ln 4} \leq f(x) \leq \frac{2(\ln 2 - 2)}{2 \ln 2} + \frac{1}{\ln 2}$$

$$\Rightarrow \frac{1}{8} + \frac{1}{4 \ln 2} \leq f(x) \leq 1 - \frac{1}{\ln 2}$$

$$2 \Rightarrow \frac{1}{8} < f(x) < 1, \forall x \in [2, 4]$$

165. Let $f: (0, 1) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{1}{1-e^{-x}}$, and $g(x) = (f(-x) - f(x))$. Consider two statements

(I) g is an increasing function in $(0, 1)$

(II) g is one-one in $(0, 1)$

Then, :

(A) Both (I) and (II) are true

(B) Only (I) is true

(C) Neither (I) nor (II) is true

(D) Only (II) is true

माना $f: (0, 1) \rightarrow \mathbb{R}$, $f(x) = \frac{1}{1-e^{-x}}$ द्वारा परिभाषित है तथा $g(x) = (f(-x) - f(x))$ है। दो कथनों का विचार कीजिए।

(I) $(0, 1)$ में g एक वर्धमान फलन है

(II) $(0, 1)$ में g एकैकी है

तो, :

(A) (I) तथा (II) दोनों सत्य है

(B) केवल (I) सत्य है

(C) न तो (I) न ही (II) सत्य है

(D) केवल (II) सत्य है

Ans. A

Sol. $g(x) = f(-x) - f(x) = \frac{1}{1-e^x} - \frac{1}{1-e^{-x}}$

$$\Rightarrow g(x) = \frac{1+e^x}{1-e^x}$$

$$g'(x) = \frac{(1-e^x)(e^x) - (1+e^x)(-e^x)}{(1-e^x)^2}$$

$$g'(x) = \frac{2e^x}{(1-e^x)^2} > 0$$

$g(x)$ is increasing as well as one-one.

166. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \frac{x^2 + 2x + 1}{x^2 + 1}$. Then

(A) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$

(B) $f(x)$ is one-one in $(-\infty, \infty)$

(C) $f(x)$ is many-one in $(1, \infty)$

(D) $f(x)$ is many-one in $(-\infty, -1)$

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ एक फलन $f(x) = \frac{x^2 + 2x + 1}{x^2 + 1}$ है। तब

(A) $[1, \infty)$ में $f(x)$ एकैकी है परन्तु $(-\infty, \infty)$ में एकैकी नहीं है।

(B) $(-\infty, \infty)$ में $f(x)$ एकैकी है।

(C) $(1, \infty)$ में $f(x)$ बहु-एक है।

(D) $(-\infty, -1)$ में $f(x)$ बहु-एक है।

Ans. A

Sol. $\because f(x) = \frac{(x+1)^2}{(x+1)}$

$$\therefore f'(x) = \frac{(x^2+1) \cdot 2(x+1) - (x+1)^2(2x)}{(x^2+1)^2} = -\frac{2(x+1)(x-1)}{(x^2+1)^2}$$

clearly $f(x)$ is one – one is $(-\infty, -1)$ and also in $(1, \infty)$ but $f(x)$ is not one – one is $(-\infty, \infty)$

167. Let f and g be twice differentiable functions on $\mathbb{R}$ such that

$$f''(x) = g''(x) + 6x, \quad f'(1) = 4g'(1) - 3 = 9, \quad f(2) = 3g(2) = 12$$

Then which of the following is NOT true?

(A) $g(-2) - f(-2) = 20$

(B) If $-1 < x < 2$, then $|f(x) - g(x)| < 8$

(C) There exists $x_0 \in (1, 3/2)$ such that $f(x_0) = g(x_0)$

(D) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

माना फलन f तथा g , $\mathbb{R}$ पर दो बार अवकलनीय हैं तथा

$$f''(x) = g''(x) + 6x, \quad f'(1) = 4g'(1) - 3 = 9, \quad f(2) = 3g(2) = 12$$

हैं। तो निम्न में से कौन सा सत्य नहीं है ?

(A) $g(-2) - f(-2) = 20$ है

(B) यदि $-1 < x < 2$ है, तो $|f(x) - g(x)| < 8$ है

(C) $x_0 \in (1, 3/2)$ का अस्तित्व है, जिसके लिए $f(x_0) = g(x_0)$ है

(D) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

Ans. B

Sol. $f''(x) = g''(x) + 6x$

$$\Rightarrow \int f''(x) dx = \int g''(x) dx + \int 6x dx + C$$

$$\Rightarrow f'(x) = g'(x) + 3x^2 + C \quad \dots\dots(1)$$

$$x = 1 \Rightarrow f'(1) = g'(1) + 3 + C$$

$$\Rightarrow 9 = 3 + 3 + C \Rightarrow C = 3$$

$$\text{from (1)} \quad \int f'(x)dx = \int g'(x)dx + \int 3x^2 dx + \int 3dx + k$$

$$\Rightarrow f(x) = g(x) + x^3 + 3x + K$$

$$x = 2 \Rightarrow f(2) = g(2) + 8 + 6 + K$$

$$\Rightarrow 12 = 4 + 8 + 6 + K$$

$$\Rightarrow K = -6$$

$$\Rightarrow f(x) = g(x) + x^3 + 3x - 6$$

$$\text{Let } h(x) = f(x) - g(x) = x^3 + 3x - 6$$

$$\Rightarrow h'(x) = f'(x) - g'(x) = 3x^2 + 3 > 0 \quad \forall x$$

$$\Rightarrow h'(x) \text{ is S.I.} \Rightarrow h(x) = 0 \text{ has no solution}$$

$$\Rightarrow h(x) \in (h(-1), h(2)) \Rightarrow h(x) \in (-8, 8)$$

$$|f'(x) - g'(x)| < 6 \Rightarrow |3x^2 + 3| < 6 \Rightarrow 3x^2 + 3 < 6$$

$$\Rightarrow x^2 - 1 < 0 \Rightarrow x \in (-1, 1)$$

$$h(-2) = f(-2) - g(-2) = -20$$

168. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function such that $f(0) = 0$, $f(1) = 1$, $f(2) = -1$, $f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeroes of $(3f'f'' + ff''')(x)$ is _____.

माना त्रिअवकलनीय एक फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार है कि $f(0) = 0$, $f(1) = 1$, $f(2) = -1$, $f(3) = 2$ है, तथा $f(4) = -2$, तो $(3f'f'' + ff''')(x)$ के न्यूनतम शून्यों की संख्या है _____

Ans. 5

Sol. $\because f: \mathbb{R} \rightarrow \mathbb{R}$ and $f(0) = 0$, $f(1) = 1$, $f(2) = -1$, $f(3) = 2$ and $f(4) = -2$ then $f(x)$ has atleast 4 real roots.

Then $f'(x)$ has atleast 3 real roots and $f''(x)$ has atleast 2 real roots.

$$\frac{d}{dx}(f^3 \cdot f'') = 3f^2 \cdot f' \cdot f'' + f^3 \cdot f'''$$

$$= f^2(3f' \cdot f'' + f \cdot f''')$$

Here $f^3 \cdot f''$ has atleast 6 roots.

Then its differentiation has atleast 5 distinct roots.

169. Let the set of all values of p , for which $f(x) = (p^2 - 6p + 8)(\sin^2 2x - \cos^2 2x) + 2(2 - p)x + 7$ does not have any critical point, be the interval (a, b) . Then $16ab$ is equal to _____.

माना p के सभी मानों, जिनके लिए $f(x) = (p^2 - 6p + 8)(\sin^2 2x - \cos^2 2x) + 2(2 - p)x + 7$ का एक भी क्रांतिक बिन्दु नहीं है, का समुच्चय, अंतराल (a, b) है। तो $16ab$ बराबर है _____.

Ans. 252

Sol. $f(x) = -(p^2 - 6p + 8) \cos 4x + 2(2-p)x + 7$

$$f'(x) = +4(p^2 - 6p + 8) \sin 4x + (4 - 2p) \neq 0$$

$$\sin 4x \neq \frac{2p-4}{4(p-4)(p-2)}$$

$$\sin 4x \neq \frac{2(p-2)}{4(p-4)(p-2)}$$

$$p \neq 2$$

$$\sin 4x \neq \frac{1}{2(p-4)}$$

$$\Rightarrow \left| \frac{1}{2(p-4)} \right| > 1$$

on solving we get

$$\therefore p \in \left(\frac{7}{2}, \frac{9}{2} \right)$$

$$\text{Hence } a = \frac{7}{2}, b = \frac{9}{2}$$

$$\therefore 16ab = 252$$

170. Let $f(x) = 2^x - x^2$, $x \in \mathbb{R}$. If m and n are respectively the number of points at which the curves $y = f(x)$ and $y = f'(x)$ intersects the x -axis, then the value of $m + n$ is _____.

माना $f(x) = 2^x - x^2$, $x \in \mathbb{R}$ है। यदि वक्र $y = f(x)$, $y = f'(x)$, x -अक्ष को क्रमशः m तथा n बिन्दुओं पर काटते हैं, तो $m + n$ का मान है _____.

Ans. 5

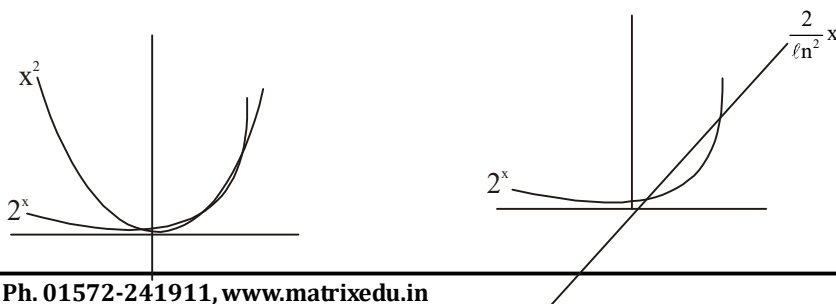
Sol. $f(x) = 2^x - x^2$

$$f(x) = 0 \quad \Rightarrow 2^x = x^2$$

$$m = 3$$

$$f'(x) = 2^x \ln 2 - 2x = 0$$

$$2^x = \frac{2}{\ln 2} x$$



$$n = 2$$

$$m + n = 5$$

171. The number of points, where the curve $y = x^5 - 20x^3 + 50x + 2$ crosses the x-axis, is _____.

उन बिंदुओं, जहाँ वक्र $y = x^5 - 20x^3 + 50x + 2$, x-अक्ष को काटता है, की संख्या है _____

Ans. 5

Sol. $y = x^5 - 20x^3 + 50x + 2$

$$\frac{dy}{dx} = 5x^4 - 60x^2 + 50 = 5(x^4 - 12x^2 + 10)$$

$$\frac{dy}{dx} = 0 \Rightarrow x^4 - 12x^2 + 10 = 0$$

$$\Rightarrow x^2 = \frac{12 \pm \sqrt{144 - 40}}{2}$$

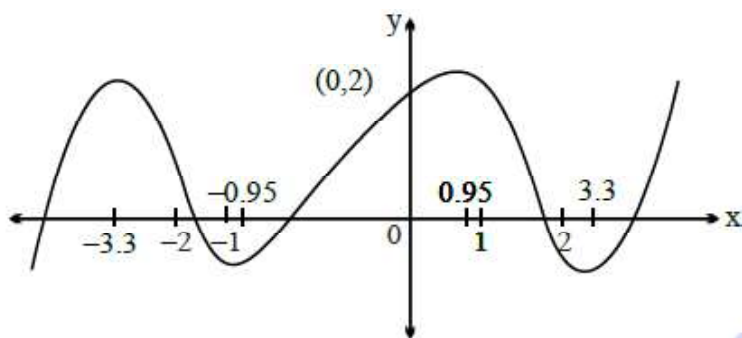
$$\Rightarrow x^2 = 6 \pm \sqrt{26} \Rightarrow x^2 \approx 6 \pm 5.1$$

$$\Rightarrow x^2 \approx 11.1, 0.9$$

$$\Rightarrow x \approx \pm 3.3, \pm 0.95$$

$$f(0) = 2, f(1) = +ve, f(2) = -ve$$

$$f(-1) = -ve, f(-2) = +ve$$



172. If a_α is the greatest term in the sequence $a_n = \frac{n^3}{n^4 + 147}$, $n = 1, 2, 3, \dots$ then α is equal to _____.

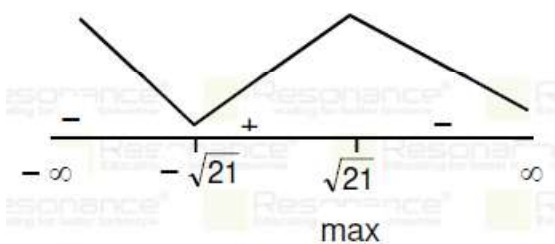
यदि अनुक्रम $a_n = \frac{n^3}{n^4 + 147}$, $n = 1, 2, 3, \dots$ का अधिकतम पद a_α है, तो α बराबर है _____

Ans. 5

Sol.
$$f'(n) = \frac{3n^2(n^4 + 147) - 4n^3(n^3)}{(n^4 + 147)^2}$$

$$f'(n) = \frac{n^2 [3n^4 + 441 - 4n^4]}{(n^4 + 147)^2}$$

$$f'(n) = \frac{-n^2(n^4 - 441)}{(n^4 + 147)^2} = \frac{-n^2(n^2 + 21)(n^2 - 21)}{(n^4 + 147)^2} = \frac{-n^2(n^2 + 21)}{(n^4 + 147)^2} (n + \sqrt{21})(n - \sqrt{21})$$



$$\text{When } n = 4 \quad a_4 = \frac{4^3}{4^4 + 147} = \frac{64}{256 + 147} = \frac{64}{403}$$

$$\text{When } n = 5 \quad a_5 = \frac{125}{625 + 147} = \frac{125}{772}$$

$$a_5 \text{ is max} = \frac{125}{772} \text{ hence } n = 5$$

173. The number of points, where the curve $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$ cuts x-axis, is equal to _____.

उन बिंदुओं, जहाँ वक्र $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$, x-अक्ष को काटता है, की संख्या है _____

Ans. 2

Sol. $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$

$$\lim_{x \rightarrow -\infty} f(x) = 1, \quad f(0) = -3, \quad \lim_{x \rightarrow \infty} f(x) \rightarrow \infty$$

One root of $f(x) = 0$ is negative and other root of $f(x) = 0$ is positive

Curve $y = f(x)$ cuts x-axis at two points

Maxima & Minima

174. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = |x + 2| - 2|x|$. If m is the number of points of local minima and n is the number of points of local maxima of f , then $m + n$ is :

एक फलन $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x + 2| - 2|x|$ द्वारा परिभाषित है। यदि f के स्थानीय निम्निष्ठ बिन्दुओं की संख्या m और स्थानीय उच्चिष्ठ बिन्दुओं की संख्या n है, तो $m + n$ बराबर है :

(A) 3

(B) 4

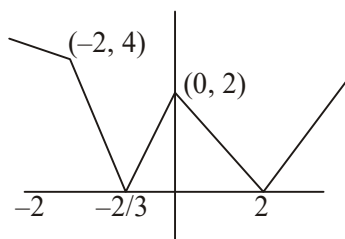
(C) 5

(D) 2

Ans. A

Sol. $f(x) = |x + 2| - 2|x|$

Critical points, $0, -2, 2, -\frac{2}{3}$



No. of maxima = 1

No. of minima = 2

175. Let $a > 0$. If the function $f(x) = 6x^3 - 45ax^2 + 108a^2x + 1$ attains its local maximum and minimum values at the points x_1 and x_2 respectively such that $x_1 x_2 = 54$, then $a + x_1 + x_2$ is equal to :

माना $a > 0$ है। यदि फलन $f(x) = 6x^3 - 45ax^2 + 108a^2x + 1$ के स्थानीय अधिकतम और न्यूनतम मान बिन्दुओं x_1 और x_2 पर प्राप्त करता है और $x_1 x_2 = 54$ है, तो $a + x_1 + x_2$ बराबर है :

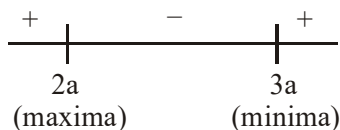
- (A) 15 (B) 13 (C) 24 (D) 18

Ans. D

Sol. $f'(x) = 18x^2 - 90ax + 108a^2$

$$= 18(x^2 - 5ax + 6a^2)$$

$$= 18(x - 2a)(x - 3a)$$



$$6a^2 = 54$$

$$a = 3$$

$$x_1 = 6, x_2 = 9$$

176. Let $x = -1$ and $x = 2$ be the critical points of the function $f(x) = x^3 + ax^2 + b \log_e |x| + 1$, $x \neq 0$. Let m and M respectively be the absolute minimum and the absolute maximum values of f in the interval $\left[-2, -\frac{1}{2}\right]$. Then $|M + m|$ is equal to (Take $\log_e 2 = 0.7$) :

माना फलन $f(x) = x^3 + ax^2 + b \log_e |x| + 1$, $x \neq 0$ के क्रान्तिक बिन्दु $x = -1$ और $x = 2$ हैं। माना अंतराल $\left[-2, -\frac{1}{2}\right]$ में f के निरपेक्ष

न्यूनतम और निरपेक्ष अधिकतम मान क्रमशः m और M है। तो $|M + m|$ बराबर है ($\log_e 2 = 0.7$ लीजिए) :

- (A) 20.9 (B) 22.1 (C) 21.1 (D) 19.8

Ans. C

Sol. $f(x) = x^2 + ax^2 + b\ln|x| + 1, x \neq 0$

$$f'(x) = 3x^2 + 2ax + \frac{b}{x}$$

$$f'(-1) = 3 - 2a - b = 0$$

$$f'(-2) = 12 + 4a - \frac{b}{2} = 0$$

$$a = \frac{-9}{2}, b = 12$$

$$f'(x) = 3x^2 - 9x + \frac{12}{x} = \frac{3(x+1)(x+2)^2}{x}$$

Max. at $x = -1$

$$f(x) = x^2 - \frac{9}{2}x^2 + 12\ln|x| + 1$$

$$f(-1) = -1 - \frac{9}{2} + 1 = -\frac{9}{2}$$

$$M = -4.5$$

Min. value at $x = -2$

$$f(-2) = -8 - 18 + 12\ln 2 + 1$$

$$m = -25 + 12\ln 2 = -16.6$$

$$|M + m| = 21.1$$

177. Let $f(x) = \int_0^{x^2} \frac{t^2 - 8t + 15}{e^t} dt, x \in \mathbb{R}$. Then the numbers of local maximum and local minimum points of f , respectively, are :

- (A) 2 and 3 (B) 1 and 3 (C) 3 and 2 (D) 2 and 2

माना $f(x) = \int_0^{x^2} \frac{t^2 - 8t + 15}{e^t} dt, x \in \mathbb{R}$ है। तब f के क्रमशः स्थानीय उच्चतम और स्थानीय निम्नतम बिन्दुओं की संख्याएँ हैं :

- (A) 2 और 3 (B) 1 और 3 (C) 3 और 2 (D) 2 और 2

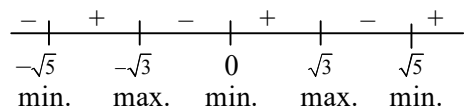
Ans. A

Sol. $f(x) = \int_0^{x^2} \frac{t^2 - 8t + 15}{e^t} dt \quad x \in \mathbb{R}$

$$f'(x) = \frac{x^4 - 8x^2 + 15}{e^{x^2}} \times 2x - 0$$

$$f'(x) = \frac{(x^2 - 3)(x^2 - 5) \times 2x}{e^{x^2}}$$

$$f'(x) = \frac{(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{5})(x + \sqrt{5}) \times 2x}{e^{x^2}}$$



No. of Maxima = 2

No. of minima = 3

178. The sum of all local minimum values of the function

$$f(x) = \begin{cases} 1 - 2x, & x < -1 \\ \frac{1}{3}(7 + 2|x|), & -1 \leq x \leq 2 \\ \frac{11}{18}(x - 4)(x - 5), & x > 2 \end{cases}$$

is :

फलन

$$f(x) = \begin{cases} 1 - 2x, & x < -1 \\ \frac{1}{3}(7 + 2|x|), & -1 \leq x \leq 2 \\ \frac{11}{18}(x - 4)(x - 5), & x > 2 \end{cases}$$

के सभी स्थानीय न्यूनतम मानों का योग है :

(A) $\frac{171}{72}$

(B) $\frac{131}{72}$

(C) $\frac{167}{72}$

(D) $\frac{157}{72}$

Ans. D

Sol.
$$f(x) = \begin{cases} 1-2x & x < -1 \\ \frac{1}{3}(7-2x) & -1 \leq x < 0 \\ \frac{1}{3}(7+2x) & 0 \leq x \leq 2 \\ \frac{11}{18}(x-4)(x-5) & x > 2 \end{cases}$$

Sum of local minimum values = $\frac{7}{3} - \frac{11}{72} = \frac{157}{72}$

179. Let $f(x) = 3\sqrt{x-2} + \sqrt{4-x}$ be a real valued function. If α and β are respectively the minimum and the maximum values of f , then $\alpha^2 + 2\beta^2$ is equal to :

माना $f(x) = 3\sqrt{x-2} + \sqrt{4-x}$ वास्तविक मूल्य फलन है। यदि α तथा β क्रमशः f के न्यूनतम तथा उच्चतम मान हैं, तो $\alpha^2 + 2\beta^2$ बराबर है :

- (A) 24 (B) 38 (C) 44 (D) 42

Ans. D

Sol. $f'(x) = \frac{3}{2\sqrt{x-2}} - \frac{1}{2\sqrt{4-x}} : f'(x) = 0 \Rightarrow x = \frac{19}{5} \in [2, 4]$

$f(2) = \sqrt{2}, f\left(\frac{19}{5}\right) = 2\sqrt{5}, f(4) = 3\sqrt{2}$

$\therefore \alpha = \sqrt{2}, \beta = 2\sqrt{5}$

$\alpha^2 + 2\beta^2 = 2 + 40 = 42$

180. Let a rectangle ABCD of sides 2 and 4 be inscribed in another PQRS such that the vertices of the rectangle ABCD lie on the sides of the rectangle PQRS. Let a and b be the sides of the rectangle PQRS when its area is maximum. Then $(a+b)^2$ is equal to :

माना भुजाओं 2 तथा 4 का एक आयत ABCD एक अन्य आयत PQRS के अंतर्गत इस प्रकार है कि आयत ABCD के शीर्ष आयत PQRS की भुजाओं पर हैं। माना आयत PQRS की भुजाएँ a तथा b है, जब इसका क्षेत्रफल है। तो $(a+b)^2$ बराबर है :

- (A) 80 (B) 72 (C) 60 (D) 64

Ans. B

Sol. Area = $(4\cos\theta + 2\sin\theta)(2\cos\theta + 4\sin\theta)$

$$= 8\cos^2 \theta + 16\sin \theta \cos \theta + 4\sin \theta \cos \theta + 8\sin^2 \theta$$

$$= 8 + 20 \sin \theta \cos \theta$$

$$= 8 + 10 \sin 2 \theta$$

$$\text{Max Area} = 8 + 10 = 18 \quad (\sin 2 \theta = 1) \quad \theta = 45^\circ$$

$$(a + b)^2 = (4\cos \theta + 2\sin \theta + 2\cos \theta + 4\sin \theta)^2$$

$$= (6\cos \theta + 6\sin \theta)^2$$

$$= 36 (\sin \theta + \cos \theta)^2$$

$$= 36 (\sqrt{2})^2$$

$$= 72$$

